

JEE MAIN REVISION MASTER: SEMICONDUCTOR SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 92 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Electronic Devices - Core Application	92 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Electronic Devices - Core Application (Very High Yield — 92 PYQs)

Core Principle: Key problem pattern in Electronic Devices.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The output V for the given inputs A and B to the circuit is :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "If X and Y are the inputs, the given circuit works as _____." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?

- (A) $A=0, B=1, C=1$
- (B) $A=1, B=0, C=0$
- (C) $A=1, B=0, C=1$ **[Correct]**
- (D) $A=1, B=1, C=0$

Correct Answer: C

Solution Analysis: Y_1 is the output of an OR gate whose inputs are A and B, $Y_1 = A + B$. Y_2 is an AND gate taking Y_1 and C as input $Y_2 = (A + B) \cdot C$. Y_3 is an AND gate taking Y_2 and A as input, $Y_3 = [(A + B) \cdot C] \cdot A$. $Y_3 = A \cdot (A + B) \cdot C = A \cdot C$. So, the truth table is,

A	B	C	$Y_1 = A + B$	$Y_2 = (A + B) \cdot C$	$Y_3 = A \cdot C$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	0
1	1	1	1	1	1

LED1 glows when $Y_1=1$ and LED2 glows when $Y_3=1$. So for both the LEDs to glow we need $Y_1=1$ and $Y_3=1$.